Problem solving with integers and decimals

Task A: Integers, negatives and decimals in the real world

1) Aisha has £13.45 credit on her school account. She spends £2.67 then spends £7.89. Aisha then puts £4.21 back into her account.

How much does Aisha have in her school account now?

$13.45 - 2.67 - 7.89 + 4.21 =$

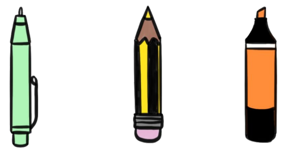
$13.45 + (-2.67) + (-7.89) + 4.21 =$

$17.66 + (-10.56) = 7.1$ Aisha has £7.10 left.

2) Jun has a £20 note. He buys 4 green pens, 6 pencils and 2 orange highlighters.

The shopkeeper gave Jun £6.16 in change.

Is this correct? Show your working out.



£1.56 76p £1.27

$4 \times 1.56 = 2 \times 2 \times 1.56 = 2 \times 3.12 = 6.24$

$6 \times 0.76 = 3 \times 0.76 \times 2 = 2.28 \times 2 = 4.56$

$2 \times 1.27 = 2.54$

$6.24 + 4.56 + 2.54 = 13.34$

$20 - 13.34 = 6.66$ The change is incorrect. Jun should get £6.66

3) There are two places in Antarctica scientists want to investigate. They look at the temperature in the morning over 5 days.

Location 1	−24.3℃	−21.6℃	−20.9℃	−24.15℃	−24.8℃

$(-24.3) + (-21.6) + (-20.9) + (-24.15) + (-24.8) = (-115.75)$

$(-115.75) \div 5 = (-115.75) \div 10 \times 2 = (-11.575) \times 2 = (-23.15)$

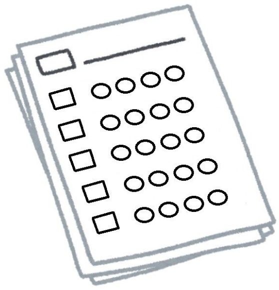
Location 2	−22℃	−21℃	−26℃	−24℃	−24.8℃

$(-22) + (-21) + (-26) + (-24) + (-24.8) = (-117.8)$

$(-117.8) \div 5 = (-117.8) \div 10 \times 2 = (-11.78) \times 2 = (-23.56)$

Location 2 is slightly colder on average than location 1.

4) Andeep is taking part in a local Maths competition.



There are 10 questions.

Each correct answer gets 8 points.

Each incorrect answer gets (-6) points.

Each question not answered gets (-2) points.

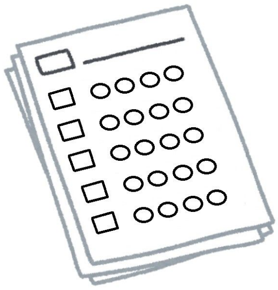
a) Andeep answers 4 correct, 3 incorrect and misses 2 questions. How many points does he have?

$4 \times 8 + 3 \times (-6) + 2 \times (-2) =$

$32 + (-18) + (-4) =$

$32 + (-22) = 10$ *Andeep has 10 points*

4b) In a different competition, Lucas gets 36 points.



There are 8 questions.

Each correct answer gets 12 points.

Each incorrect answer gets (-4) points.

Each question not answered gets (-2) points.

Give some examples of how Lucas performed in the competition.

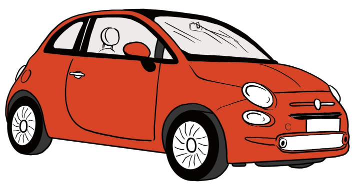
Example

$4 \times 12 + 2 \times (-4) + 2 \times (-2) = 36$

4 correct, 2 incorrect and 2 missed

Task B: Measurements and accuracy

1) Here is an example of two car rental companies.



Rent-a-Red-Car

1 km = £0.46

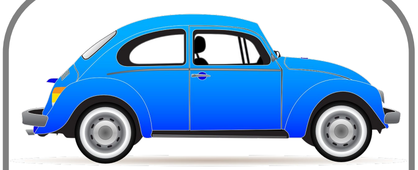
Rent-a-Red-car

0.46 × 70 =
4.6 × 7 = 32.20

Rent-a-Blue-car

2500 m = 2.5 km
70 ÷ 2.5 = 700 ÷ 25 = 28
28 × 1.1 = 28 × 11 × 0.1 = 30.8

£32.20 - £30.80 = £1.40



Rent-a-Blue-Car

2500 m = £1.10

An Oak teacher knows he will need to rent a car for a 70 km journey. Work out the difference in prices between the two companies for a 70 km car hire.

2) Sofia helps her neighbour feed her cats using a feeding chart.

Mass to the nearest 10 g	**Quantity of dry food per day**
1 - 2.99 kg	70 g
3 kg - 4.99 kg	90 g
5 - 5.99 kg	100 g
more than 6 kg	115 g

1.5 kg of dry food
£9.82

Her neighbour has three cats weighing 4.5 kg, 5.1 kg and 6.2 kg.

How much will it cost to feed all three cats for 2 weeks?

4.5 kg cat requires 90 g per day

5.1 kg cat requires 100 g per day

6.2 kg cat requires 115 g per day

Total food per day is 90 + 100 + 115 = 305 g per day

For 2 weeks it is 305 × 14 = 4 270g

1.5 kg = 1500 g

She will need 3 bags.

9.82 × 3 = £29.46 needs to be spent on cat food.